1.43 Trigonometry: Modelling in Degrees

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Sin Graph

44

What does it look like?

Key Features:
• Repeats every $360°$.
• Range: $-1 < \sin(x) < 1$

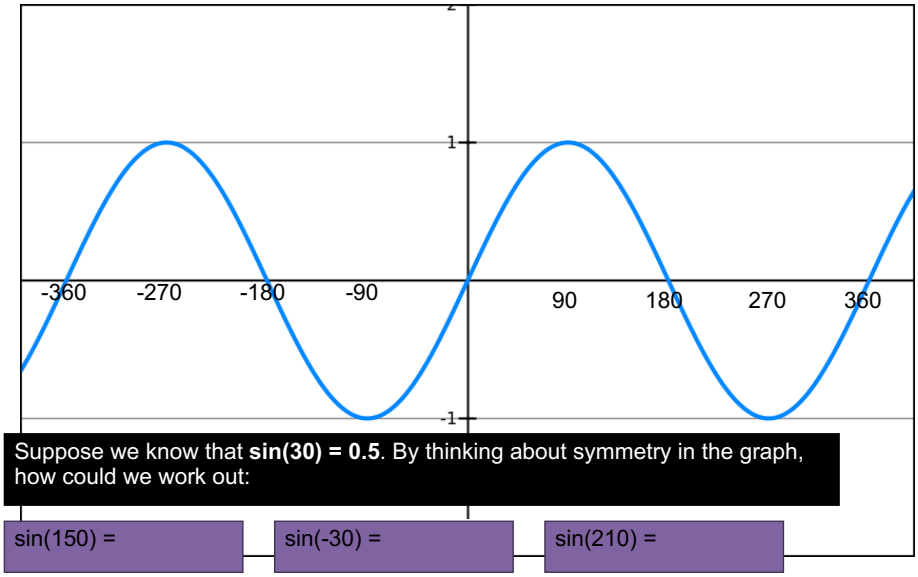
-360 -270 -180 -90 90 180 270 360

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Sin Graph

45

What do the following graphs look like?



Suppose we know that **sin(30) = 0.5**. By thinking about symmetry in the graph, how could we work out:

sin(150) = | sin(-30) = | sin(210) =

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Cos Graph

46

What do the following graphs look like?

Key Features:
• Repeats every $360°$.
• Range: $-1 < \sin(x) < 1$

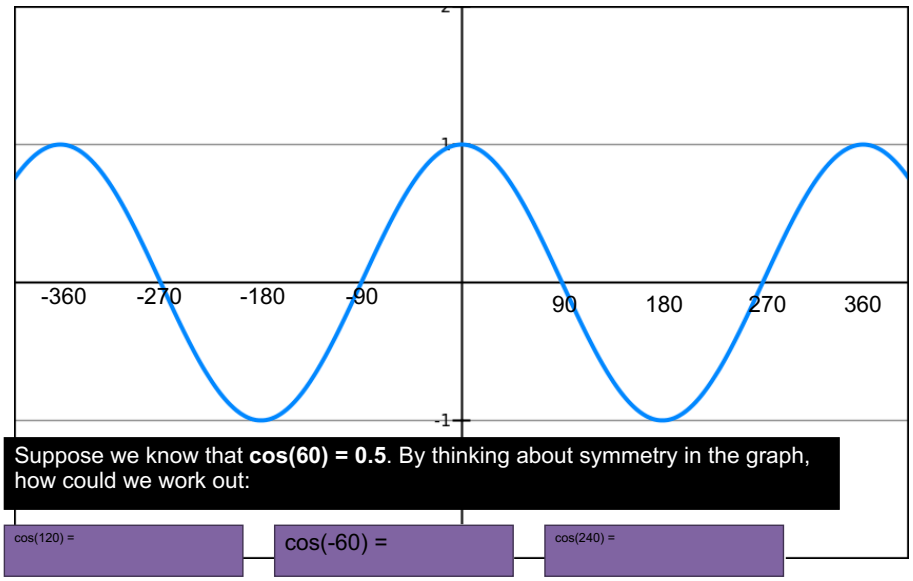
-360 -270 -180 -90 90 180 270 360

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Cos Graph

47

What does it look like?



Suppose we know that **cos(60) = 0.5**. By thinking about symmetry in the graph, how could we work out:

cos(120) =

cos(-60) =

cos(240) =

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Tan Graph

48

What does it look like?



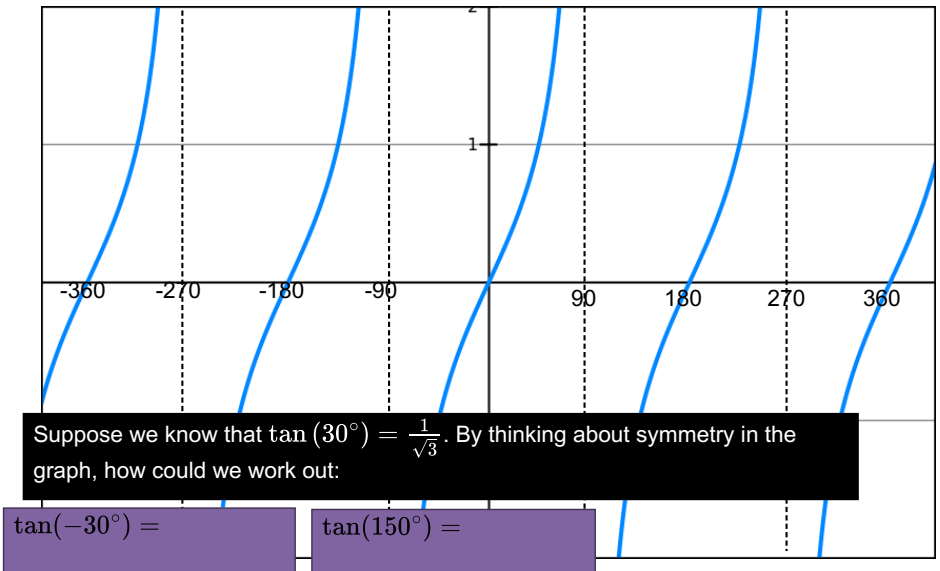
Key Features:
- Repeats every $\mathbf{180}^\circ$.
- Roots: $0, 180^\circ, -180^\circ, \ldots$
- Range: $\tan(x) \in \mathbb{R}$ (i.e. no min/max value!)
- Asymptotes: $x = \pm 90^\circ, \pm 270^\circ, \ldots$

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Tan Graph

49

What does it look like?



Suppose we know that $\tan(30^\circ) = \frac{1}{\sqrt{3}}$. By thinking about symmetry in the graph, how could we work out:

$\tan(-30^\circ) =$

$\tan(150^\circ) =$

1.43 Trigonometry: Modelling in Degrees

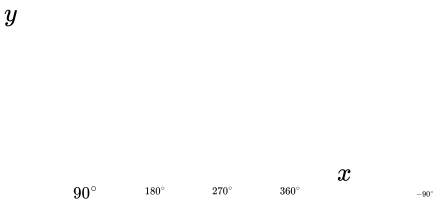
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Transforming Trigonometric Graphs

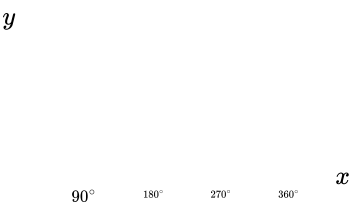
50

There is no new theory here: just use your knowledge of transforming graphs, i.e. whether the transformation occurs ‘inside’ the function (i.e. input modified) or ‘outside’ the function (i.e. output modified).

Sketch $y = 4\sin x,\ 0 \leq x \leq 360^\circ$



Sketch $y = \cos(x + 45^\circ),\ 0 \leq x \leq 360^\circ$

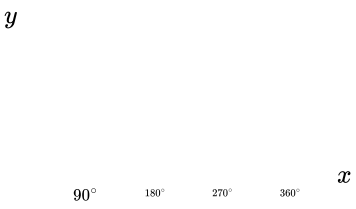


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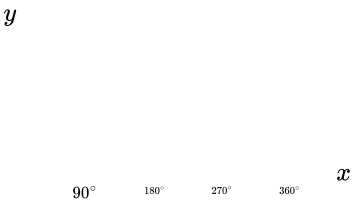
Transforming Trigonometric Graphs

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Sketch $y = -\tan x,\ 0 \leq x \leq 360^\circ$



Sketch $y = \sin\left(\frac{x}{2}\right),\ 0 \leq x \leq 360^\circ$



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Exercise 9F – Y1 Page 194

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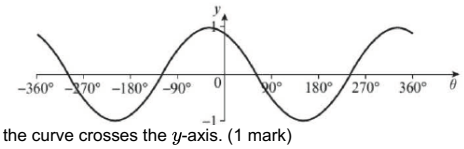
1 Sketch the graph of $y = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.\par2 Sketch the graph of $y = \tan\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.\par3 Sketch the graph of $y = \sin\theta$ in the interval $-90^\circ \leq \theta \leq 270^\circ$.

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Exercise 9G – Y1 Pages 197-198

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7 The graph shows the curve
$y = \cos(x + 30^\circ), -360^\circ \leq x \leq 360^\circ$.
a Write down the coordinates of the points where the curve crosses the x-axis. (2 marks)



b Find the coordinates of the point where the curve crosses the y-axis. (1 mark)

b $\cos 30^\circ = \frac{\sqrt{3}}{2}$; $\left(0, \frac{\sqrt{3}}{2}\right)$

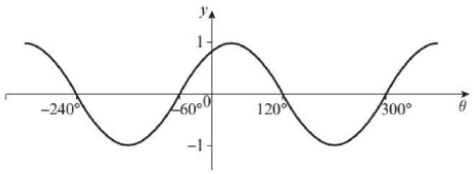
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Exercise 9G – Y1 Pages 197-198

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8 The graph shows the curve with equation $y = \sin(x + k), -360^\circ \leq x \leq 360^\circ$, where k is a constant.
a Find one possible value for k. (2 marks)
b Is there more than one possible answer to part a? Give a reason for your answer. (2 marks)



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Exercise 9G – Y1 Pages 197-198

55

9 The variation in the depth of water in a rock pool can be modelled using the function $y = \sin(30t)^\circ$, where t is the time in hours and $0 \leq t \leq 6$.
a Sketch the function for the given interval.
b If $t = 0$ represents midday, during what times will the rock pool be at least half full?